

COMPUTER ARCHITECTURE

Introduction



Administrative issues

- Instructor:
 - Assoc. Prof. Dr. Cuong Pham-Quoc (in Vietnamese Phạm Quốc Cường)
 - Computer Engineering Department, Faculty of Computer Science and Engineering, HCMUT
 - Email: cuongpham@hcmut.edu.vn
 - Homepage: www.cse.hcmut.edu.vn/~cuongpham

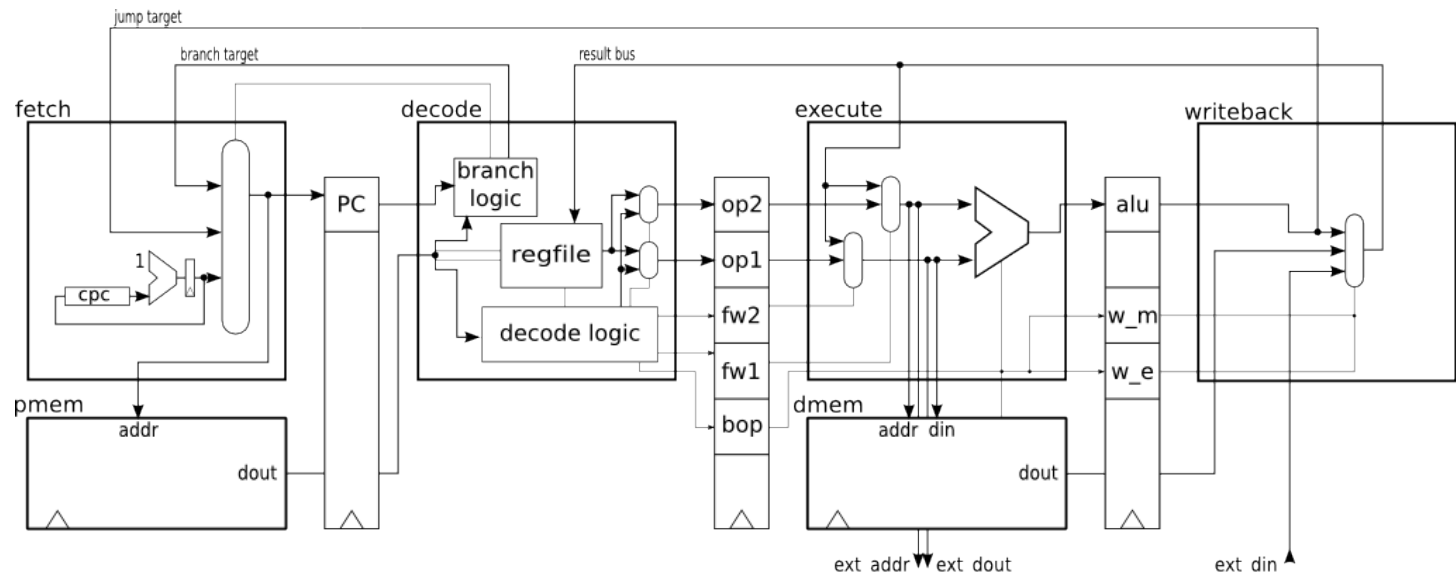
Computer

- Q: What is a Computer?
 - A: “an **electronic machine** that is used for storing, organizing, and finding words, numbers, and pictures, for doing calculations, and for controlling other machines” – Cambridge dictionary
 - A: “a **general-purpose device** that can be programmed to carry out a set of arithmetic or logical operations automatically” - Wikipedia

Abacus

Computer architecture

- Q: What is Computer Architecture?
 - A: “the **science** and **art** of selecting and interconnecting hardware components to create computers that meet **functional**, **performance** and **cost** goals” - WWW Computer Architecture Page



The course

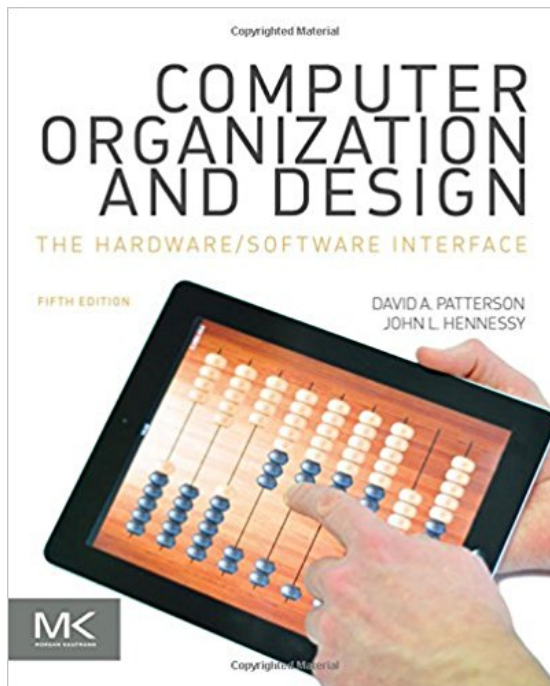
- Elementary course for both **Computer Engineering** and **Computer Science** programs
- Contents:
 - Performance evaluation
 - Instruction set architecture
 - Computer arithmetic
 - Data-path and control signals
 - Memory & I/O system
 - Multicores, Multiprocessors, and Clusters

Outcomes

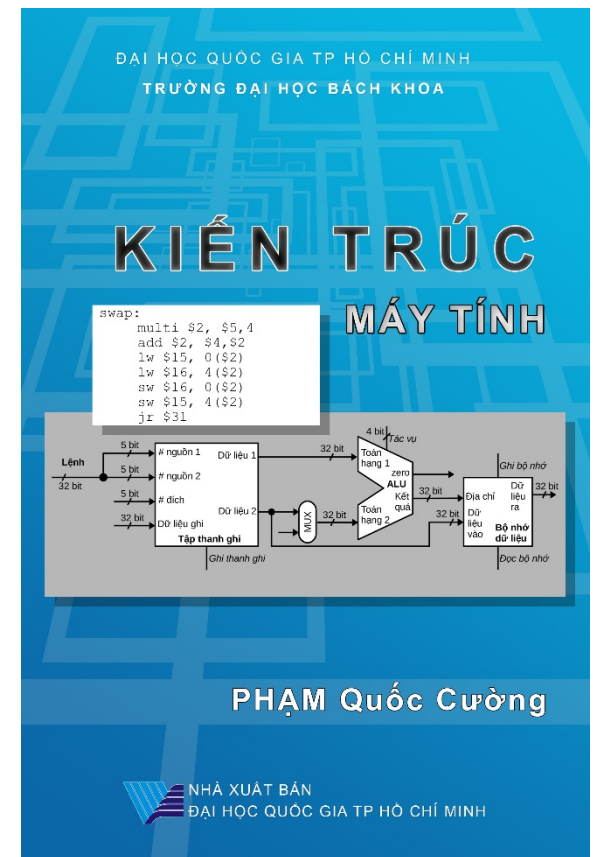
- Outcomes:
 - Understand the structure, organization of a computer system: the main components and the basic principles of its operations
 - Write and optimize small programs and fragments of codes to demonstrate an understanding of machine level operation

Learning materials

- Slides/handouts
 - BKEL
 - www.cse.hcmut.edu.vn/~cuongpham
- Textbooks



Computer Architecture (c) Cuong Pham-Quoc/HCMUT



Assessment

- Assignment: 30%
- Labs: 10% (mandatory)
- Mid-term: 20% – multiple choices/writing, with 1 A4 piece of paper note
- Final exam: 40% – multiple choices/writing, with 2 A4 piece of paper note

In-class regulations



handout

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